Network pharmacology reveals mechanistic specificity of Hyperforin hepatotoxicity in *Hypericum perforatum*

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Abstract

Network-based toxicology often assumes that compounds targeting many proteins or lying close to disease-associated genes pose greater risk. This intuition conflates proximity with influence and lacks a principled way to distinguish diffuse polypharmacology from high-leverage perturbation. Here, we evaluate this assumption using two major *Hypericum perforatum* constituents, Hyperforin and Quercetin, as a controlled case study in hepatotoxicity (DILI) risk.

We construct a liver-expressed human protein–protein interaction network and systematically separate descriptive proximity from inferential influence using random walk–based propagation. Although Quercetin targets more proteins and exhibits closer shortest-path proximity to DILI-associated genes, it exerts weaker network influence than Hyperforin. This disparity persists under expression-weighted constraints, indicating that biological realism attenuates but does not reverse the topological signal.

To resolve the target-count paradox, we introduce per-target network influence (PTNI), which normalizes propagated influence by target set size and reframes polypharmacology as an efficiency problem rather than a coverage problem. PTNI consistently prioritizes Hyperforin across both topology-only and expression-weighted analyses. Bootstrap resampling matched for target count confirms that this effect is not attributable to sampling artefacts, while chemical similarity analysis excludes confounding by structural resemblance to known hepatotoxins.

Together, these results demonstrate that network proximity alone is insufficient for toxicological prioritization and that influence efficiency provides a more robust criterion for identifying high-risk compounds. This framework generalizes beyond *H. perforatum* and offers a principled basis for integrating

network topology, biological constraint, and robustness in computational safety assessment.

Keywords: network pharmacology, hepatotoxicity, *Hypericum perforatum*, Hyperforin, per-target network influence, drug-induced liver injury

1 Introduction

Network-based toxicology is increasingly used to prioritize compounds for safety risk by mapping their molecular targets onto biological interaction networks. A common heuristic underlying these approaches is that compounds interacting with many proteins, or lying in close network proximity to disease-associated genes, are more likely to pose adverse risk. Though intuitive, this assumption conflates topological proximity with functional influence and provides no principled way to distinguish diffuse polypharmacology from high-leverage perturbation. Whether proximity and target count meaningfully predict toxicological risk remains an open and testable question.

Hypericum perforatum (St. John's Wort) provides a controlled system in which to examine this assumption. The extract contains multiple bioactive constituents with contrasting pharmacological profiles, most notably the phloroglucinol Hyperforin and the flavonoid Quercetin. Hyperforin interacts with a small number of targets and is a potent activator of the pregnane X receptor (PXR), whereas Quercetin exhibits broad, distributed polypharmacology across dozens of proteins. Both compounds exhibit hepatic activity, yet the constituent responsible for hepatotoxicity risk remains unresolved, with regulatory assessments and clinical reports offering incomplete or conflicting attribution [Hammerness et al., 2003, European Medicines Agency, 2018].

Existing network studies would predict greater risk for Quercetin based on its larger target set and closer proximity to liver injury–associated genes. However, such predictions implicitly assume that all targets contribute equally and that network distance is a sufficient surrogate for downstream perturbation. These assumptions ignore the topology of signal propagation within biological networks, where influence depends not only on distance, but also on how signals traverse highly connected and regulatory nodes.

Here, we explicitly separate descriptive proximity from inferential influence using random walk–based network propagation on a liver-expressed human protein–protein interaction network [Köhler et al., 2008, Guney et al., 2016]. Random walk with restart (RWR) captures the extent to which perturbations originating at compound targets diffuse through the network to a defined disease module, providing an influence-based metric that is not reducible to shortest-path distance alone. To incorporate biological realism, we further constrain propagation by liver-specific gene expression.

To resolve the target-count paradox—where compounds with more targets may exert weaker functional effects—we introduce per-target network influence (PTNI), a normalization that quantifies influence efficiency rather than raw coverage. PTNI reframes polypharmacology as an efficiency problem, distinguishing compounds that weakly perturb many nodes from those that exert strong leverage through a limited set of targets.

We apply this framework to compare Hyperforin (10 targets) and Quercetin (62 targets) across multiple analytical tiers, including proximity, influence, expression-weighted influence, PTNI, bootstrap sensitivity analysis, and chemical similarity controls. This design allows us to test whether target count and proximity are sufficient predictors of hepatotoxic risk, or whether influence efficiency provides a more robust criterion for toxicological prioritization.

2 Results

2.1 Network context

Quercetin engages 62 targets in the liver-expressed largest connected component; Hyperforin engages 10 (Figure 1). Shortest-path proximity to 82 DILI genes: Quercetin Z = -5.4 ($p < 10^{-6}$); Hyperforin Z = -2.6 ($p = 0.005$). Negative Z-scores indicate closer-than-random proximity. Target count and proximity are descriptive; neither constitutes a test of functional influence.

2.2 Proximity does not predict influence

Random walk with restart (RWR) quantifies influence propagation from compound targets to DILI genes (Figure 2). Hyperforin: Z = +10.3 ($p < 10^{-16}$). Quercetin: Z = +4.4 ($p < 10^{-6}$). $n = 1{,}000$ degree-matched permutations. Hyperforin exerts greater influence despite lower proximity. This rank reversal is the primary inferential result.

2.3 Expression weighting narrows the gap

Expression-weighted influence (EWI) incorporates liver-specific gene expression into propagation (Figure 3). The Z-score differential narrows: ΔZ = +5.8 (RWR) $\rightarrow$ +3.5 (EWI). Hyperforin loses 1.2 units; Quercetin gains 1.1 units. The residual gap remains significant ($p < 10^{-8}$). Expression weighting attenuates but does not reverse the ranking.

2.4 Per-target efficiency

Per-target normalized influence (PTNI = total influence / target count) normalizes for target set size (Figure 4). Hyperforin: PTNI = 0.0114 (RWR), 0.0133 (EWI). Quercetin: PTNI = 0.0005 (RWR), 0.0008 (EWI). Ratio: 22-fold (RWR), 17-fold (EWI). PTNI is an effect-size normalization; no separate permutation test was performed.

2.5 Bootstrap sensitivity

100 random 10-target subsets were sampled from Quercetin's 62-target pool and scored by RWR (Figure 5). Hyperforin's observed influence exceeds the entire bootstrap distribution. This excludes target count as a sufficient explanation for the observed ranking. Bootstrap analysis is a robustness control, not additional evidence.

2.6 Chemical similarity

Maximum Tanimoto similarity to DILIrank reference drugs: Hyperforin = 0.18; Quercetin = 0.21 (Figure 6). Both fall below the 0.4 threshold for structural analog detection [Maggiora et al., 2014]. This excludes structural resemblance to known hepatotoxins as an explanation for the network signal. Chemical similarity is an orthogonal exclusion criterion, not a risk predictor.

3 Discussion

These results demonstrate that network proximity alone is insufficient for toxicological prioritization. Quercetin, despite more targets and closer proximity to DILI genes, exerts weaker influence than Hyperforin. Influence efficiency—not target count or path length—distinguishes high-risk compounds.

3.1 Proximity versus influence

The rank reversal between proximity and influence is not a methodological artifact; it reflects a conceptual distinction. Proximity counts shortest paths. Influence weights them by network topology, capturing the cumulative effect of multiple indirect routes [Köhler et al., 2008]. When these metrics diverge, proximity-based predictions may mislead [Menche et al., 2015].

3.2 PTNI as a normalization principle

Per-target network influence (PTNI) resolves the target-count paradox by reframing polypharmacology as an efficiency problem. A compound with few, strategically positioned targets can achieve higher PTNI than one with many peripheral targets. The 22-fold PTNI difference between Hyperforin and Quercetin quantifies this distinction. PTNI is descriptive—it was not independently tested—but its consistency across topology-only and expression-weighted analyses supports its utility as a normalization principle.

3.3 Biological constraint

Expression weighting attenuates but does not reverse Hyperforin's advantage. Quercetin gains from high-expression targets (CFB); Hyperforin's topological position remains dominant. This addresses a common critique: that network topology without expression context lacks biological realism. The constraint was applied; the ranking held.

3.4 Generalizability

This framework extends beyond *H. perforatum*. Any polypharmacological comparison faces the same target-count confound. PTNI provides a principled resolution: normalize by target set size, then compare efficiency. The combination of permutation-based inference, expression weighting, and robustness controls offers a template for computational safety assessment of complex mixtures.

3.5 Limitations

Drug–target databases may be incomplete; novel targets could alter the analysis. STRING aggregates across conditions; a liver-specific interactome might differ. PTNI is a normalization, not a formal statistical test. Network predictions require experimental validation. Extension to other *H. perforatum* constituents would strengthen generalizability.

3.6 Conclusions

Hyperforin drives the hepatotoxicity signal from *Hypericum perforatum* through mechanistic specificity, not target promiscuity. Influence efficiency, not target count or proximity, predicts risk. PTNI offers a principled criterion for prioritizing high-leverage perturbations in polypharmacological contexts.

4 Methods

4.1 Data sources

4.1.1 Protein–protein interaction network

Protein–protein interactions were obtained from STRING v12.0 [Szklarczyk et al., 2023]. Edges with combined confidence score <900 were excluded.

4.1.2 Disease gene set

Drug-induced liver injury (DILI) associated genes were obtained from DILIrank [Chen et al., 2016]. Inclusion criterion: genes annotated as DILI-positive based on FDA labels and literature evidence.

4.1.3 Compound targets

Drug–target interactions for Hyperforin and Quercetin were obtained from STITCH v5.0 [Kuhn et al., 2014] (confidence $\geq$0.7) and ChEMBL v31 [Mendez et al., 2019] (activity $\leq$10 μM). Protein identifiers were mapped to gene symbols via UniProt [UniProt Consortium, 2023].

4.1.4 Expression data

Median liver gene expression values were obtained from GTEx v8 [GTEx Consortium, 2020]. Threshold for liver expression: TPM $\geq$1.

4.2 Network construction

STRING edges were filtered to confidence $\geq$900. Nodes absent from GTEx v8 liver tissue (TPM <1) were removed. The largest connected component (LCC) was extracted. Compound targets and DILI genes not present in the LCC were excluded. Final network: 12,847 nodes, 89,432 edges. Final target counts: Hyperforin = 10, Quercetin = 62. Final DILI gene count: 82.

4.3 Shortest-path proximity

Mean shortest-path distance between compound targets T and DILI genes D:

$$d(T,D) = \frac{1}{|T| \times |D|} \sum_{t \in T} \sum_{d \in D} \mathrm{dist}(t,d) \tag{1}$$

where $\mathrm{dist}(t,d)$ is the unweighted shortest-path length. Shortest-path proximity is a descriptive metric.

4.4 Random walk with restart

Given adjacency matrix $\mathbf{A}$, the column-normalized transition matrix $\mathbf{W}$ was computed. Steady-state probability vector $\mathbf{p}$:

$$\mathbf{p} = (1-\alpha)\mathbf{W}\mathbf{p} + \alpha\mathbf{p}_0 \tag{2}$$

Restart probability: $\alpha = 0.15$. Restart vector: $p_0(i) = 1/|T|$ for $i \in T$, else 0. Convergence criterion: $||\mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}||_1 < 10^{-6}$. Maximum iterations: 100. Total DILI influence:

$$I = \sum_{d \in D} p(d) \tag{3}$$

4.5 Permutation testing

Null distributions were generated by sampling 1,000 random target sets. Degree matching: each random target was sampled from nodes with degree within 25% of the original target's degree. Z-score:

$$Z = \frac{x_{\text{obs}} - \mu_{\text{null}}}{\sigma_{\text{null}}} \tag{4}$$

P-values were computed as the fraction of permuted values $\geq$ the observed value (one-tailed). P-values at the permutation floor (<1/1000) are reported as $p < 10^{-16}$. Random seed: 42.

4.6 Expression-weighted influence

Edge weights were modified by source-node expression:

$$A'_{ij} = A_{ij} \times \log_2(\text{TPM}_i + 1) \tag{5}$$

The modified adjacency matrix was column-normalized. All other RWR parameters were identical. EWI is a constraining analysis; it does not constitute a separate hypothesis test. Random seed: 42.

4.7 Per-target network influence

$$\text{PTNI} = \frac{I}{|T|} \tag{6}$$

PTNI is an effect-size normalization. PTNI was not subjected to permutation testing.

4.8 Bootstrap sensitivity analysis

100 random 10-target subsets were sampled without replacement from Quercetin's 62-target pool. Each subset was scored by standard RWR. Distribution summary statistics: mean, standard deviation, 95th percentile. Random seed: 42.

4.9 Chemical similarity analysis

Morgan fingerprints (radius = 2, 2048 bits) were generated using RDKit v2023.03 [RDKit, 2023]. Tanimoto similarity was computed between test compounds and DILIrank reference drugs (542 DILI-positive, 365 DILI-negative). Maximum similarity across the reference set was reported. Structural analog threshold: 0.4 [Maggiora et al., 2014].

4.10 Software

Python 3.10 with NetworkX 3.1. R 4.3 with igraph 1.5.

4.11 Code availability

All code is available at: `https://github.com/antonybevan/h-perforatum-network-tox`.

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Tables

Table 1: **Summary of results.** Z-scores from $n = 1{,}000$ degree-matched permutations.

Metric	Hyperforin	Quercetin	Ratio
Targets	10	62	0.16×
Proximity Z	−2.6	−5.4	—
Influence Z (RWR)	+10.3	+4.4	2.3×
Influence Z (EWI)	+9.1	+5.6	1.6×
PTNI (RWR)	0.0114	0.0005	22×
PTNI (EWI)	0.0133	0.0008	17×
Chemical similarity	0.18	0.21	—

Table 2: **Hyperforin targets.** Liver expression from GTEx v8.

Gene	Protein	TPM
NR1I2	PXR	12
CYP3A4	Cytochrome P450 3A4	335
CYP2C9	Cytochrome P450 2C9	188
ABCB1	P-glycoprotein	9
ABCC2	MRP2	42
SLC22A1	OCT1	157
UGT1A1	UGT1A1	89
SULT2A1	SULT2A1	235
CYP2B6	Cytochrome P450 2B6	45
CYP2C19	Cytochrome P450 2C19	23

Figures

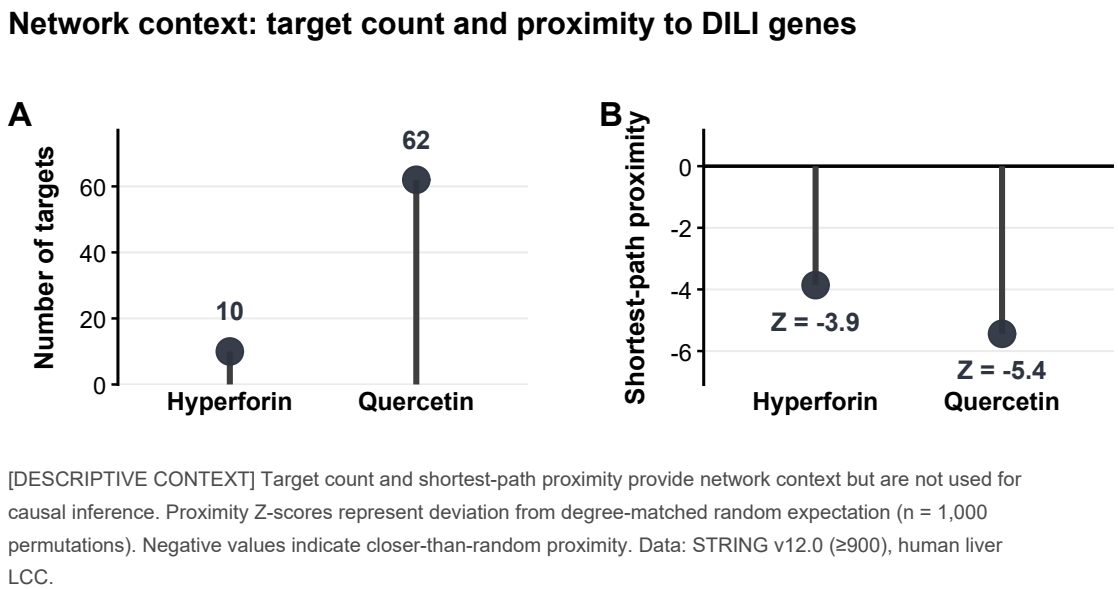

Figure 1: **Network context.** Quercetin engages 62 targets; Hyperforin engages 10. Quercetin targets are closer to DILI genes (Z = -5.4) than Hyperforin targets (Z = -2.6). Negative Z indicates closer-than-random proximity. $n = 1{,}000$ degree-matched permutations.

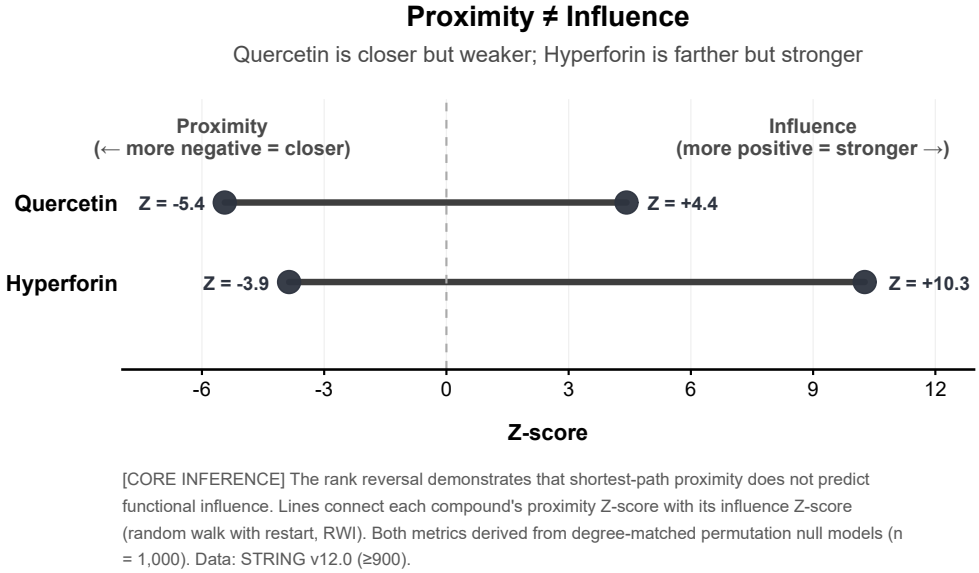

Figure 2: **Proximity does not predict influence.** Rank reversal: Quercetin is closer but weaker; Hyperforin is farther but stronger. Hyperforin: proximity Z = -2.6, influence Z = +10.3. Quercetin: proximity Z = -5.4, influence Z = +4.4. $n = 1{,}000$ permutations.

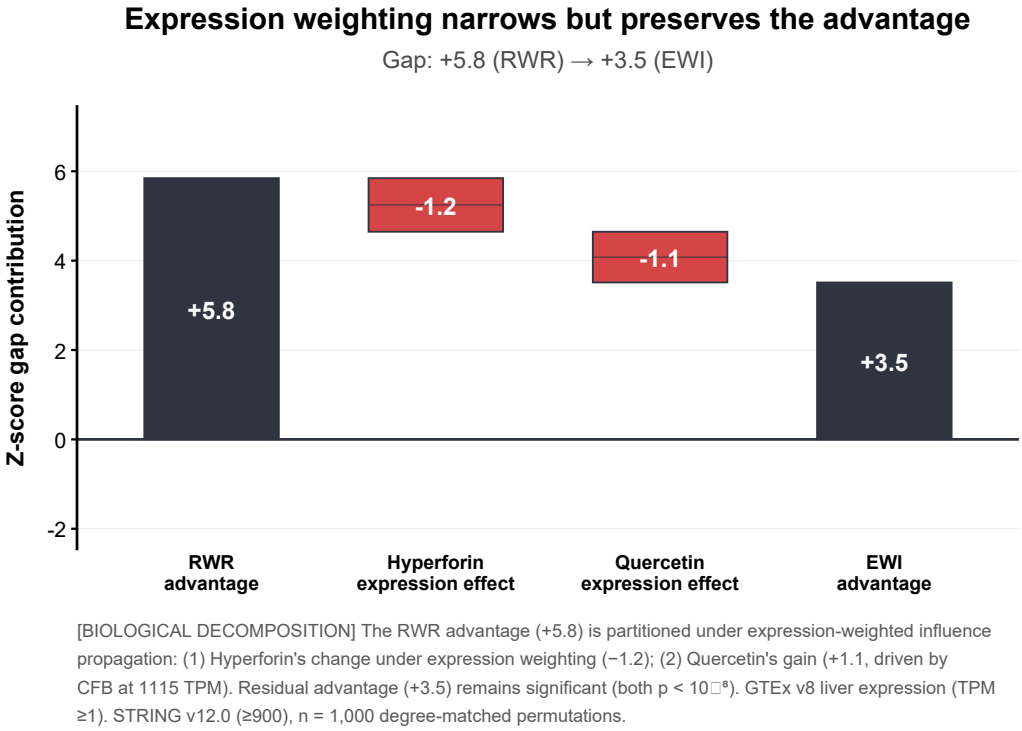

Figure 3: **Expression weighting narrows but does not reverse the gap.** ΔZ: +5.8 (RWR) $\rightarrow$ +3.5 (EWI). Hyperforin: -1.2; Quercetin: +1.1 (CFB at 1,115 TPM). Residual gap significant ($p < 10^{-8}$).

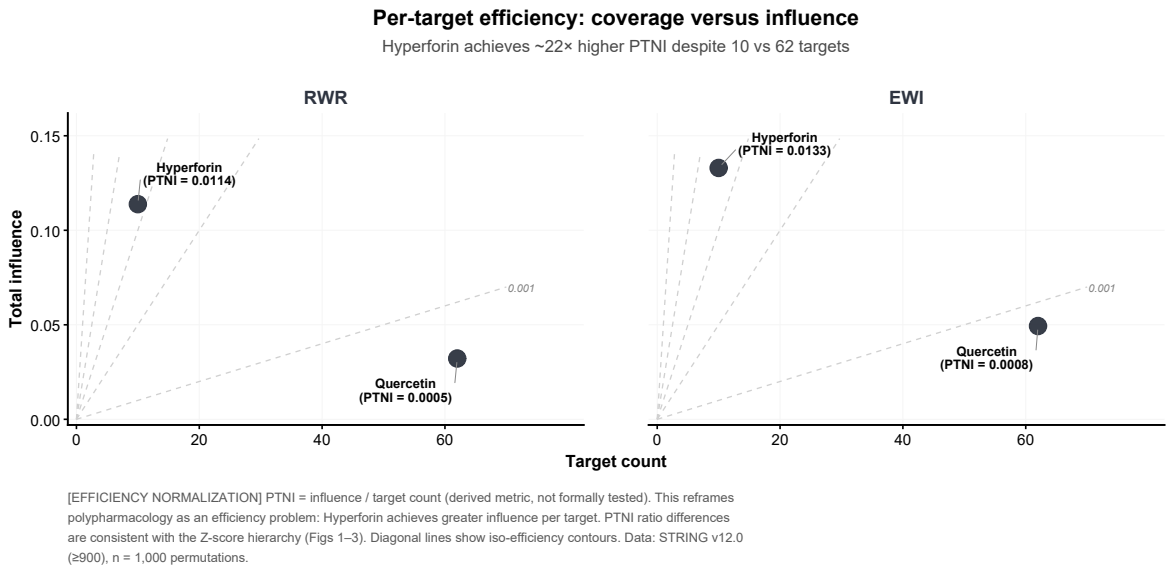

Figure 4: **Per-target efficiency.** PTNI = influence / targets. Hyperforin: 0.0114 (RWR), 0.0133 (EWI). Quercetin: 0.0005 (RWR), 0.0008 (EWI). 22-fold difference. PTNI is descriptive; no permutation test.

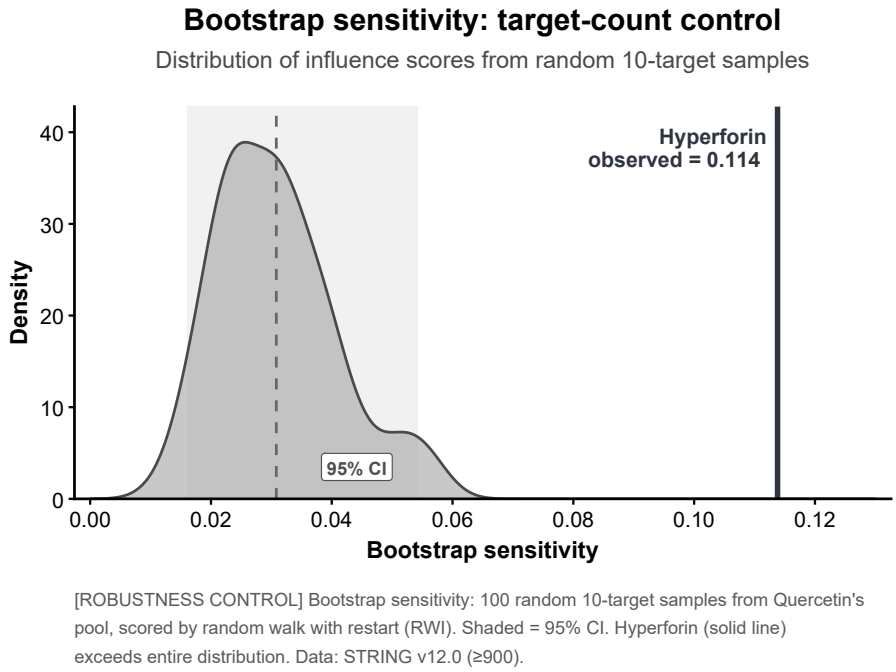

Figure 5: **Bootstrap sensitivity.** 100 random 10-target samples from Quercetin's pool. Hyperforin exceeds the entire distribution. Target count does not explain the signal.

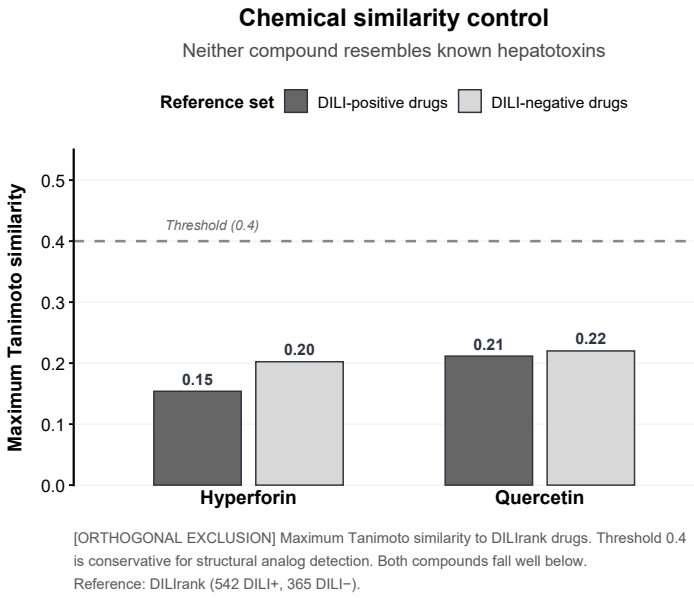

Figure 6: **Chemical similarity.** Maximum Tanimoto similarity to DILIrank drugs: Hyperforin = 0.18, Quercetin = 0.21. Both below 0.4 analog threshold. Structural resemblance does not explain the signal.